VNIVERSIDAD D SALAMANCA

Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial

Curso para las Facultades de Ciencias y de Ciencias Químicas

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ÍNDICE DE CONTENIDOS

Bienvenido a **Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial**.

¿Qué es esto?

Es el material del curso y una plantilla diseñada para que el profesorado de las **Facultades de Ciencias y de Ciencias Químicas de la USAL** pueda crear libros docentes interactivos de forma sencilla. La idea principal de esta plantilla es que sea lo suficientemente extensa para cubrir muchos casos de uso y que cada alumno la adapte a su propio curso, pero que a la vez esté lo suficientemente equipada para que se pueda usar de forma sencilla con la ayuda de asistentes de IA (como GitHub Copilot, Gemini, Claude, Codex, etc.) y con un editor de código como VS Code.

Contenido

En este libro encontrarás:

- Tutoriales para aprender a usar la plantilla
- Ejemplos por Grado para ver casos reales
- Información sobre cómo citar y licencias

Versión PDF

También puedes descargar la versión imprimible del libro:

- Descargar PDF en español
- Download PDF in English

> **Note**
>
> Este proyecto está diseñado para ser usado con **VS Code** y asistentes de **IA**.

Part I

Matemáticas

CAPÍTULO 1

ECUACIONES DE SEGUNDO GRADO

Las ecuaciones de segundo grado tienen la forma general:

$$ax^2 + bx + c = 0 \tag{1.1}$$

Donde a, b y c son constantes y $a \neq 0$.

1.1 Fórmula general

La solución a la ecuación (1.1) viene dada por la fórmula general:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Part II

Informática

EJERCICIOS SEMANA 3: PROGRAMACIÓN III

CONTENIDOS: Wrapper Classes y primeros ejercicios de POO

2.1 Ejercicios obligatorios:

1. Deberás crear una clase `Person.java` con atributos nombre, altura y peso, sus correspondientes getters y setters. En el método main deberás instanciar 3 objetos de dicha clase y completar su información solicitándola al usuario por consola para cada objeto. Finalmente, deberás mostrar por consola quién es el más alto y quién es el que más pesa. Deberás utilizar siempre los getters y setters para obtener la información y establecerla en cada instancia de cada objeto.
2. Mejora el ejercicio 1 para que exista un constructor sin parámetros que inicialice los valores de nombre, altura y peso a valores por defecto que estimes oportunos.
3. Mejora el ejercicio 1 de tal forma que la clase `Person` ofrezca un método que calcule el IMC de la persona a través de los valores actuales de los atributos del objeto (su estado). Se deberá contemplar la posibilidad de que los valores no se hayan inicializado a valores válidos.
4. Analiza el flujo de ejecución del programa colocando puntos de interrupción en el constructor y en las distintas partes del código del ejercicio 1. Establece valores a los atributos directamente en su declaración y establece un punto de parada para comprobar qué se ejecuta antes: el constructor o la declaración del atributo.

2.2 Ejercicios de extensión:

5. Empleando los métodos `parseXXX` de las clases de envoltorio, construir una clase dotada de métodos adecuados para leer todos los tipos de datos numéricos (`int`, `float`, `double`) en forma de cadenas. Si el usuario no inserta un valor correcto, pedir de nuevo el valor.
6. Modifica el ejercicio 1 para que la clase `Person.java` esté dentro de un paquete llamado `model` en lugar del mismo paquete que la clase `App.java`. ¿Qué has necesitado cambiar para poder utilizarla en `App.java`?
7. ¿Qué diferencia existe entre el constructor por defecto de una clase en Java y un constructor sin parámetros?
8. ¿Por qué utilizar getters y setters y no acceder directamente a los atributos a través del punto? Encapsulación. ¿Es una cuestión de seguridad o de diseño?
9. ¿Qué diferencia hay entre definir un atributo como `public`, `private` o no poner modificador de visibilidad? ¿En qué caso es posible acceder a ellos mediante el punto (.) con una referencia de un objeto?
10. ¿Para qué se utiliza la keyword `this`? ¿Y la keyword `new`?

11. Es posible definir clases que a su vez tengan atributos de otras clases (denominado Composición). Debes ahora modificar el ejercicio 1 para que la clase `Person` tenga un atributo de tipo `DNI` (otra clase que deberéis crear) que tendrá como atributos número y letra. Al constructor de la clase `Person` se le deberá pasar como parámetro un número de DNI y se creará un objeto de tipo `DNI`, generando la letra correspondiente a partir del algoritmo para obtener la letra del DNI a partir del número. Consulta el operador `%` (módulo, resto tras la división entera) en Java.
12. Cree una clase llamada `Rectangulo.java` con los atributos longitud y anchura, cada uno con un valor predeterminado de 1. Debe tener métodos para calcular el perímetro y el área del rectángulo. Debe tener métodos getter y setter para longitud y anchura. Los métodos setter deben verificar que longitud y anchura sean números de punto flotante mayores de 0.0, y menores de 20.0. Escriba un programa para probar la clase `Rectangulo`.
13. Cree un nuevo proyecto con la clase `Person.java` que creó inicialmente. Realice las siguientes instrucciones y responda a las preguntas planteadas.

```java
public static void main(String[] args) {
    // Creación de un objeto de la clase Person
    Person paco = new Person("Paco", 22, 177.3f, 82.3f);

    // Copiamos la referencia del objeto anterior a otra referencia
    Person refPaco2 = paco;

    // Utilizamos la segunda referencia para establecer un valor en el objeto al que
↪hace referencia
    refPaco2.setAgeYears(77);

    // ¿Resultado?
    System.out.printf("La edad de %s es:%d%n", paco.getName(), paco.getAgeYears());

    // ¿Cuál es la edad que se muestra?
    // ¿Cuántos objetos hay en memoria? ¿Cuántas referencias? Revisa el Debugger
}
```

Part III

Investigación

SISTEMA MULTIAGENTE PARA EVALUACIÓN CERVICAL

Este capítulo se basa en el artículo *“A novel multiagent system for cervical motor control evaluation and individualized therapy: integrating gamification and portable solutions”* [MBR+24].

3.1 Resumen del estudio

El estudio se centró en diseñar un dispositivo portátil y objetivo para evaluar y tratar las alteraciones del Control Motor Cervical (CMC). Este dispositivo se basa en una arquitectura propuesta que emplea tecnología avanzada para evaluar y mejorar el CMC de los pacientes.

Durante un estudio piloto con 10 participantes, se verificó la viabilidad y usabilidad del dispositivo, incluyendo una evaluación inicial utilizando el *Head Relocation Test* y una intervención de 12 sesiones a lo largo de 4 semanas.

La arquitectura del sistema propuesto se encarga de recopilar datos pertinentes sobre el control motor cervical de los pacientes. Emplea algoritmos avanzados para procesar estos datos y evaluar objetivamente la función del CMC. Además, el sistema adapta la terapia a las necesidades individuales de cada paciente.

Los resultados preliminares indican que el dispositivo y la arquitectura propuesta impactan positivamente en la precisión del rendimiento de las pruebas de evaluación. Si bien se requieren pruebas de validación adicionales para confirmar su eficacia, este dispositivo surge como una alternativa prometedora y valiosa para evaluar y tratar a pacientes con alteraciones del CMC. Su enfoque en tecnología avanzada y adaptación personalizada se alinea con investigaciones previas en telerrehabilitación.

3.2 Arquitectura del Sistema

El sistema utiliza un entorno multiagente que coordina el flujo de información y personaliza los ejercicios terapéuticos. Para ello, incorpora sensores en miniatura (*Inertial Measurement Units*, IMU) que recogen métricas de movimiento en tiempo real, corrigiendo desviaciones mediante algoritmos de fusión de datos.

3.3 Referencias del artículo

Las referencias completas del estudio son: [AbdollahiMKuberPMShiraishiMSoangraRRashediE, AiQLiuZMengWLiuQXieSQ, BarraOrtizHAMatamalaAMInostrozaFLArayaCLMondacaVN, BarzegarKhanghahAFernieGRFekrA, BisioIGaribottoCLavagettoFSciarroneA, BlasHSSMendesASEncinasFGSilvaLAGonzalezGV, CalvaresiDMarinoniMDragoniAFHilfikerRSchumacherM, ChenJOrCKChenT, ChenKBSestoMEPontoKLeonardJMasonAVanderheidenGWilliamsJRadwinRG, DePauwJMercelisRHallemansAMichielsSTruijenSCrasPDHertoghW, DonSDKooningMVoogtLIckmansKDaenenLNijsJ, DuQBaiHZhuZ, GrazioliETranchitaEBorrielloGCerulliCMingantiCParisiA, GuidanceM, HidalgoPerezAFGarciaALdUraldeVillanuevaIGMartinezAPAlemanyAFernandezCarneroJLToucheR, HumayunMJhanjhiNZAlmotilagAAlmufarehMF, KazeminasabSNejadghaderiSAAmiriPPourfathiHAKhodaeiMSullmanMJKolahiAASafiriS, KheraPKumarN, KimWSChoSKuJKimYLeeKHwangHJPaikNJ, KumarSRavellaUKShrivastavaAKAnilMSwamySK, LaranjoLDunnAGTongHLKocaballiABChenJBashirRSurianDGallegoBMagrabiFLauAYeal, MalflietAKregelJCoppietersIPauwRDMeeusMRousselNCagnieBDanneelsLNijsJ, MichielsSHertoghWDTruijenSNovemberDWuytsFHeyningPV, MihajlovicZPopovicSBrkicKCosicK, ParkJSJungYJLeeG, PengBYangLLiYLiuTLiuYa, PengBYangLLiYLiuTLiuYb, QuNTianHDMartinoEZhangB, QuNTianHMartinoEDZhangB, RiffittsMOhAAlemuAPatelVSmithCNMuratiSBailesAAllenMDombrowskiMLeeJYeal, RodaCRodriguezACLJaqueroVNavarroEGonzalezP, SpencerJWolfSLKesarTM, SremakaewMJullGTreleavenJUthaikhupS, SulisEMarianiSMontagnaS, TreleavenJa, TreleavenJb, VinoskiS, WeizmanYTiroshOFussFKTanAMRutzE, YoonTLKimHNMinJH, ZoeteRMJOsmotherlyPGRivettDASnodgrassSJ]

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Part IV

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